

Mohammad Taha Majlesi

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PROFILE & RESEARCH INTERESTS

Passionate Computer Engineering student at the University of Tehran with a strong research focus on **Deep Generative Models**, **Large Language Models (LLMs)**, **Deep Reinforcement Learning**, and cognitive-inspired **System 2 AI**. Extensive experience as a Research Assistant in explainable AI and financial modeling, and as a Teaching Assistant for over 5 graduate and undergraduate courses across top universities. Committed to open-source contribution, with all research code, datasets, and model artifacts publicly available on GitHub and Hugging Face.

EDUCATION

University of Tehran

Bachelor of Science in Computer Engineering

– **GPA:** 18.0 / 20

– **Relevant Coursework (Graduate & Undergraduate):** Deep Generative Models, Large Language Models, Deep Reinforcement Learning (Sharif Univ), System 2 AI, Machine Learning, Computer Vision, Data Science.

Tehran, Iran

Sep 2022 – Present

RESEARCH EXPERIENCE

Tehran Institute for Advanced Studies (TeIAS)

Research Assistant

- Conducted research on **Large Language Model (LLM) Explainability**, investigating internal attention mechanisms to improve transparency in transformer-based architectures.
- Developed advanced predictive models for cryptocurrency markets using time-series analysis and machine learning techniques.
- Contributed to the design and deployment of AI development tools to streamline academic research workflows.

Tehran, Iran

Jul 2025 – Feb 2026

Robust and Interpretable Machine Learning Lab

Research Assistant

- Conducted research under the supervision of **Dr. Rohban** and **Dr. Jafari**, focusing on robustness and interpretability in deep learning models.

Tehran, Iran

Jun 2025 – Oct 2025

TEACHING EXPERIENCE

Sharif University of Technology

Teaching Assistant

- **Compiler Programming:** Assisted in course delivery, designing assignments related to lexical analysis, parsing, and code generation.
- **Introduction to Programming with Python:** Mentored students under the supervision of **Dr. Zarei**, conducting code reviews and troubleshooting sessions.

Tehran, Iran

Oct 2025 – Present

University of Tehran

Teaching Assistant (Multiple Courses)

- **Artificial Intelligence:** TA for Prof. Yaghoob Zadeh and Dr. Fadaei (Oct 2025 – Present).
- **Machine Learning (Graduate Course):** TA for the Master's level course under **Dr. Aboalghasemi** and **Dr. Arrabi** (Sep 2025 – Oct 2025).
- **Advanced Programming (C++):** Lead TA under **Prof. Ramtin Khosravi**. Designed complex algorithmic assignments and developed an **automated grading agent** in Python to evaluate student submissions against test cases.
- **Introduction to Computing Systems & Programming:** TA under **Prof. Hadi Moradi** and **Prof. Mahmoud Reza Hashemi**.
- **Fundamentals of Programming:** TA under **Prof. Mohammad Javad Dousti**.

Tehran, Iran

Sep 2023 – Present

PROFESSIONAL EXPERIENCE

Quera

Django Software Engineer Intern

- Engineered and maintained robust **Django** web applications, implementing scalable RESTful APIs using **Django REST Framework (DRF)**.

Tehran, Iran

Jul 2023 – Dec 2023

- Utilized **Celery** and **Redis** for asynchronous task processing to enhance application performance.
- Deployed applications using Docker and Nginx, ensuring consistent environments across development and production.
- Contributed to the development of bootcamp training materials, mentoring new developers in Python and Django best practices.

GENERATIVE MODELING & TRUSTWORTHY AI

- **Deep Generative Models Series:** Implemented state-of-the-art architectures including **Diffusion Models (DDPM/Score-based)**, **Flow-based Models (RealNVP)**, **GANs**, and **Variational Inference** systems for the University of Tehran graduate course.
- **Unlearning from VAE:** Implemented **Selective Amnesia**, a principled approach for targeted machine unlearning in Conditional VAEs, enabling the forgetting of specific classes or features without retraining.
- **FaithScore Upgrade:** Built an evaluation framework for **Vision-Language Models (VLM)** hallucination detection using multi-prompt ensemble judging and embedding similarity.

LARGE LANGUAGE MODELS & REASONING

- **System 2 AI Reasoning:** Developed reasoning agents using **ReAct**, **CoT**, and **Reflexion** for multi-hop QA. Implemented **GRPO-based Fine-tuning** for LLMs on the GSM8k dataset.
- **Consciousness-inspired Attention:** Developed Transformer modifications based on **Global Workspace Theory (Baars)** to explore cognitive bottlenecks in LLMs.
- **LLM Bitcoin Prediction & Recommender:** Developed a private research project utilizing LLM-based embeddings and sentiment analysis for financial market forecasting.

ARTIFICIAL INTELLIGENCE & PROBABILISTIC MODELING

- **Probabilistic Graphical Models:** Implemented **Bayesian Networks** and **Hidden Markov Models (HMM)** with mathematical formulations for complex sequence modeling tasks.
- **NLP Toolkit for Persian:** Developed tools for **Hkasre Correction** and sentiment analysis on Persian datasets (Ganjoor), including drug name extraction using specialized NLP pipelines.
- **Optimization & Unsupervised Learning:** Projects covering **PCA**, **Clustering (K-Means)**, and neural network optimization strategies for model evaluation.

SYSTEMS & COMPILERS

- **SimpleLang Compiler:** Developed a multi-phase compiler (Phase 2) in Java, focusing on advanced type analysis, semantic checking, and code generation for the PLC course.
- **AP Grader Agent:** A Python-based automation system that handles multi-practice grading (A1-A6) for C++ programming courses with automated test case execution.
- **ARM Pipeline Processor:** A complete 5-stage pipeline ARM processor implementation in Verilog.

TECHNICAL SKILLS

Languages: Python (Advanced), C, C++, Java, Go, SystemVerilog, Verilog, SQL.

AI & Machine Learning: PyTorch, TensorFlow, Scikit-Learn, Transformers, LLMs, Deep Reinforcement Learning, Generative Models, Computer Vision, MLOps, LangChain.

Web Development: Django, Django REST Framework, HTML, CSS, React, GraphQL.

DevOps & Tools: Docker, Kubernetes, Git, GitHub Actions, Linux, CI/CD, PostgreSQL, Nginx, Redis, Celery.

CERTIFICATIONS

MLOps Specialization – Duke University (Sep 2024)

Fundamentals of Reinforcement Learning – University of Alberta (Sep 2024)

Machine Learning Specialization – Stanford Online (Aug 2023)

Data Analyst Career Guide – IBM (Sep 2024)

Software Engineering with Django – Quera (Dec 2023)